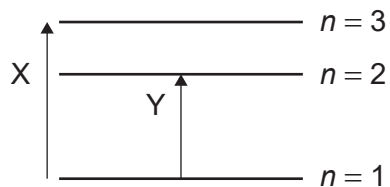


E.1 / PAPER 1 / QUESTION 27

5 November 2025 / TZ1 (file label) / HL Paper 1A / original Q33 / 8825-6707

27. Some energy levels for a hydrogen atom are shown.

diagram not to scale



What is $\frac{\text{energy absorbed in transition X}}{\text{energy absorbed in transition Y}}$?

A. $\frac{32}{27}$

B. $\frac{3}{2}$

C. 2

D. $\frac{9}{4}$